

# Locally Convex Topological Vector Spaces

## Definitions and Examples

**Definition:** A topological vector space (TVS) is a vector space  $X$  with a topology  $\mathcal{T}$  such that, with respect to this topology, the maps

- (i)  $X \times X \rightarrow X, (x, y) \mapsto x + y$
- (ii)  $\mathbb{F} \times X \rightarrow X, (\alpha, x) \mapsto \alpha x$

are continuous with respect to  $\mathcal{T}$ .

**Example:** If  $\mathcal{P}$  is a family of seminorms on  $X$ , and  $\mathcal{T}$  is the topology on  $X$  with subbasic sets  $U(x_0, p, \varepsilon) = \{x \in X : p(x - x_0) < \varepsilon\}$ , then  $(X, \mathcal{T})$  is a topological vector space.

**Definition:** A locally convex topological vector space (LCS) is a TVS equipped with a family of seminorms  $\mathcal{P}$  that satisfy

$$\bigcap_{p \in \mathcal{P}} \{x : p(x) = 0\} = \{0\}.$$

That is, the family  $\mathcal{P}$  separates the points of  $X$ .

**Proposition:** Let  $X$  be a TVS, and let  $p$  be a seminorm on  $X$ . The following are equivalent:

- (a)  $p$  is continuous;
- (b)  $\{x \in X : p(x) < 1\}$  is open;
- (c)  $0 \in \{x \in X : p(x) < 1\}^\circ$ ;
- (d)  $0 \in \{x \in X : p(x) \leq 1\}^\circ$ ;
- (e)  $p$  is continuous at 0;
- (f) there is a continuous seminorm  $q$  on  $X$  such that  $p \leq q$ .

*Proof.* We see immediately that (a) implies (b) since the set is equal to the inverse image of the open sub-set  $[0, 1) \subseteq [0, \infty)$ . Similarly, if  $p^{-1}([0, 1))$  is open, then it is equal to its own interior, and since  $0 \in p^{-1}([0, 1))$ , it follows that there is a subbasic open neighborhood of 0 contained in  $p^{-1}([0, 1))$ , giving (c). Similarly, we obtain (d) by the fact that  $0 \in p^{-1}([0, 1))^\circ \subseteq p^{-1}([0, 1])^\circ$ .

Next, we see that (d) implies that for every  $\varepsilon > 0$ ,  $0 \in p^{-1}([0, \varepsilon])^\circ$ , so if  $(x_i)_i$  is a net in  $X$  converging to 0 and  $\varepsilon > 0$ , there is some  $i_0$  such that for all  $i \geq i_0$ ,  $x_i \in \{x : p(x) \leq \varepsilon\}$ . That is,  $p(x_i) \leq \varepsilon$  for all  $i \geq i_0$ , whence  $p$  is continuous at 0. This gives (e).

If  $p$  is continuous at 0, then if  $x_i \rightarrow x$  is a convergent net in  $X$ , the reverse triangle inequality gives

$$|p(x) - p(x_i)| \leq p(x_i - x),$$

whence  $p(x_i) \rightarrow p(x)$ , so  $p$  is continuous. This gives that (e) implies (a).

Taking  $q = p$  yields (a) implies (f). We will show that (f) implies (e). If  $x_i \rightarrow 0$  in  $X$ , then  $q(x_i) \rightarrow 0$ , but since  $0 \leq p(x_i) \leq q(x_i)$ , it follows that  $p(x_i) \rightarrow 0$ .  $\square$

**Example:**

- (a) Let  $X$  be a completely regular topological space, and let  $C(X)$  be the set of all continuous functions from  $X$  to  $\mathbb{F}$ . If  $K$  is compact in  $X$ , we may define a seminorm  $p_K(f) = \sup_{x \in K} |f(x)|$ . Then,

$$\{p_K : K \subseteq X \text{ compact}\}$$

forms a separating family of seminorms for  $C(X)$ .

- (b) Let  $G$  be an open subset of  $\mathbb{C}$ . If  $H(G)$  is the set of holomorphic functions on  $G$ , then the same semi-

norms  $\{p_K : K \subseteq G \text{ compact}\}$  forms a locally compact space whose topology is uniform convergence on compact subsets.

- (c) Let  $X$  be a normed vector space. For each  $\varphi \in X^*$ , let  $p_\varphi(x) = |\varphi(x)|$ . Then,  $\mathcal{P} = \{p_\varphi : \varphi \in X^*\}$  forms a separating family of seminorms for  $X$ . The topology these seminorms induce is called the *weak topology* on  $X$ . It is often denoted  $\sigma(X, X^*)$ .
- (d) If  $X$  is a normed space, then for each  $x \in X$ , we may define a seminorm  $p_x : X^* \rightarrow [0, \infty)$  by taking  $p_x(\varphi) = |\varphi(x)|$ . This defines a separating family of seminorms  $\mathcal{P} = \{p_x : x \in X\}$ , and the topology on  $X^*$  induced by these seminorms is called the *weak\**, or  $w^*$ , topology. It is often denoted  $\sigma(X^*, X)$ .

Next, we focus on an examination of the “convexity” part of the notion of locally convex topological vector spaces.

**Definition:** A subset  $A \subseteq X$  is said to be convex if for all  $v, w \in A$ , the segment

$$[v, w] := \{tv + (1-t)w : 0 \leq t \leq 1\}$$

is contained in  $A$ .

Note that the intersection of any family of convex subsets is convex.

**Definition:** If  $A \subseteq X$  is any subset, then the *convex hull* of  $A$ , denoted  $\text{conv}(A)$ , is the intersection of all the convex subsets that contain  $A$ . If  $X$  is a TVS, the *closed convex hull* of  $A$ , denoted  $\overline{\text{conv}}(A)$ , is the intersection of all the closed convex sets that contain  $A$ .

Given any linear map  $T : X \rightarrow Y$  of real vector spaces, and  $C \subseteq X$  a convex subset of  $Y$ , then  $T^{-1}(C)$  is convex in  $X$ . If  $X^*$  is the dual of any complex vector space, then since  $X$  is also a real vector space, it then follows that the sets

$$\begin{aligned} A_0 &= \{x : |f(x)| \leq 1\} \\ A_1 &= \{x : \text{Re}(f(x)) \leq 1\} \\ A_3 &= \{x : \text{Re}(f(x)) > 1\} \end{aligned}$$

are all convex.

**Proposition:** Let  $X$  be a TVS, and let  $A$  be a convex subset of  $X$ . Then,

- (a)  $\overline{A}$  is convex;
- (b) if  $a \in A^\circ$  and  $b \in \overline{A}$ , then

$$[a, b] := \{tb + (1-t)a : 0 \leq t < 1\}$$

is contained in  $A^\circ$ .

*Proof.* Let  $a \in A$ ,  $b \in \overline{A}$ , and let  $0 \leq t \leq 1$ . Let  $(x_i)_i$  be a net in  $A$  such that  $x_i \rightarrow b$ . Then,  $tx_i + (1-t)a \rightarrow tb + (1-t)a$ . In particular, this means that if  $b \in \overline{A}$  and  $a \in A$ , then  $[a, b] \subseteq \overline{A}$ . So, given any  $a, b \in \overline{A}$ , we have that the net  $(x_i)_i \rightarrow b$  is such that  $[x_i, a] \subseteq \overline{A}$ , so it follows that  $[b, a] \subseteq \overline{A}$ , whence  $\overline{A}$  is convex.

For (b), fix  $0 < t < 1$ , and set  $c = tb + (1-t)a$  for some  $a \in A^\circ$  and  $b \in \overline{A}$ . Since  $a \in A^\circ$ , there is an open neighborhood  $V$  of 0 such that  $a + V \subseteq A$ . Therefore, for any  $d \in A$ , we have

$$\begin{aligned} A &\supseteq td + (1-t)(a + V) \\ &= t(d - b) + tb + (1-t)(a + V) \\ &= (t(d - b) + (1-t)V) + c. \end{aligned}$$

We hope to find some  $d \in A$  such that  $0 \in t(d - b) + (1-t)V$ , which we call  $U$ . The preceding inclusion then shows that  $c \in A^\circ$ , since  $U$  is an open neighborhood of 0 with  $c + U \subseteq A$ .

Finding such a  $d$  is equivalent to finding  $d$  such that  $0 \in \frac{1}{t}(1-t)V + (d - b)$ , or  $d \in b - \frac{1}{t}(1-t)V$ . Yet, we know that  $0 \in -\frac{1}{t}(1-t)V$  by the assumption that  $V$  is an open neighborhood of 0. Since  $b \in \overline{A}$ , it follows that  $b - \frac{1}{t}(1-t)V \cap A \neq \emptyset$  by the definition of closure, whence such a  $d \in A$  exists.  $\square$

**Corollary:** If  $A \subseteq X$ , then  $\overline{\text{conv}}(A) = \overline{\text{conv}(A)}$ .

**Definition:** A set  $A \subseteq X$  is called balanced if for any  $|\alpha| \leq 1$  and any  $x \in A$ , we have  $\alpha x \in A$ .

We say a set  $A$  is *absorbing* if for any  $x \in X$ , there is some  $\varepsilon > 0$  such that  $tx \in A$  for all  $0 \leq t < \varepsilon$ . We say  $A$  is absorbing at  $a$  if for any  $x \in X$ , there is  $\varepsilon > 0$  such that  $a + tx \in A$  for all  $0 \leq t < \varepsilon$ .

Given a seminorm  $p$  on  $V$ , the set  $V = \{x : p(x) < 1\}$  is a convex balanced set that is absorbing at each of its points. We will show that the converse actually holds — that is, a vector space is a LCS if and only if the set of all open, convex, balanced subsets of  $X$  forms a neighborhood basis at 0.

**Proposition:** Suppose  $X$  is a vector space over  $F$ , and  $V$  is a nonempty convex balanced set that is absorbing at each of its points. Then, there is a unique seminorm  $p$  on  $X$  such that

$$V = \{x \in X : p(x) < 1\}.$$

*Proof.* Let

$$p(x) = \inf\{t \geq 0 : x \in tV\}.$$

Since  $V$  is absorbing, it follows that  $X = \bigcup_{n=1}^{\infty} nV$ , so this is a well-defined quantity. Furthermore,  $p(0) = 0$ .

To show that  $p(\alpha x) = |\alpha|p(x)$ , we suppose  $\alpha \neq 0$ , and since  $V$  is balanced,

$$\begin{aligned} p(\alpha x) &= \inf\{t \geq 0 : \alpha x \in tV\} \\ &= \inf\left\{t \geq 0 : x \in \frac{t}{\alpha}V\right\} \\ &= \inf\left\{t \geq 0 : x \in \frac{t}{|\alpha|}V\right\} \\ &= |\alpha| \inf\left\{\frac{t}{|\alpha|} : x \in \frac{t}{|\alpha|}V\right\} \\ &= |\alpha|p(x). \end{aligned}$$

We now observe that if  $\alpha, \beta \geq 0$ , then for any  $a, b \in V$ , we have

$$\begin{aligned} \alpha a + \beta b &= (\alpha + \beta) \left( \frac{\alpha}{\alpha + \beta} a + \frac{\beta}{\alpha + \beta} b \right) \\ &\in (\alpha + \beta)V, \end{aligned}$$

since  $V$  is convex. For any  $x, y \in X$ , take  $p(x) = \alpha$  and  $p(y) = \beta$ , and let  $\delta > 0$ . Then,  $x \in (\alpha + \delta)V$  and  $y \in (\beta + \delta)V$  by the definition of the infimum, so that  $x + y \in (\alpha + \beta + 2\delta)V$ . Therefore, we have  $p(x + y) \leq \alpha + \beta + 2\delta$ , which holds for all  $\delta > 0$ , so that  $p(x + y) \leq p(x) + p(y)$ .

Finally, we will have to show that  $V = \{x : p(x) < 1\}$ . If  $p(x) = \alpha < 1$ , then for any  $\alpha < \beta < 1$ , we have  $x \in \beta V \subseteq V$ , which emerges from the fact that  $V$  is balanced. Thus,  $V \supseteq \{x : p(x) < 1\}$ . If  $x \in V$ , then  $p(x) \leq 1$ , and since  $V$  is absorbing at  $x$ , there is some  $\varepsilon > 0$  such that for  $0 < t < \varepsilon$ , we have  $y = x + tx \in V$ . Yet, this means  $x = \frac{1}{1+t}y$  with  $y \in V$ , so that  $p(x) = \frac{1}{1+t}p(y) \leq \frac{1}{1+t} < 1$ .  $\square$

The seminorm  $p$  is called the Minkowski function for  $V$ , or the *gauge* of  $V$ .

**Proposition:** Let  $X$  be a TVS, and let  $\mathcal{U}$  be the collection of open, convex, balanced subsets of  $X$ . Then,  $X$  is locally convex if and only if  $\mathcal{U}$  is a basis for the neighborhood system at 0.

*Proof.* Suppose  $X$  is locally convex with topology defined by the family of seminorms  $\mathcal{P}$ . Then, we observe that each of the subbasic open neighborhoods of 0,

$$\{U(0, p, \varepsilon) : \varepsilon > 0, p \in \mathcal{P}\},$$

are open, convex, and balanced; since the set of all finite intersections of all these subbasic open neighbor-

hoods yields a neighborhood basis for 0, and finite intersections of open, balanced, convex subsets are open, balanced, and convex, so it follows that  $\mathcal{U}$  contains the set of all finite intersections of all these subbasic open neighborhoods, whence  $\mathcal{U}$  contains a basis for the neighborhood system at 0. In particular, this means  $\mathcal{U}$  also must generate the neighborhood system at 0, so  $\mathcal{U}$  forms a basis for the neighborhood system at 0.

Now, suppose  $\mathcal{U}$  forms a basis for the neighborhood system at 0. Observe that an open, convex, balanced subset of  $V$  contains 0, and is itself absorbing. We claim that the family of Minkowski functionals

$$\mathcal{P} = \{p_V(\cdot - z) : z \in X, V \in \mathcal{U}\}$$

generates the topology for  $X$ . □

Thus, it follows that in a locally convex space, a net  $(x_i)_i$  converges in the topology to  $x$  if and only if for all seminorms  $p \in \mathcal{P}$ , we have  $p(x_i - x) \rightarrow 0$ .

**Example:**

- (i) Suppose  $X$  is a locally compact Hausdorff space (or any Tychonoff space, but we will work with these for uniformity), and let  $C(X)$  be the space of all continuous functions on  $X$ . Then, the family of seminorms  $p_K(f) = \sup_{x \in K} |f(x)|$  yields a locally convex space where convergence is defined as uniform convergence on compact subsets.
- (ii) If  $X$  is a normed space, then the family of seminorms  $p_\varphi(x) = |\varphi(x)|$  for each  $\varphi \in X^*$  defines the weak topology on  $X$ .
- (iii) If  $X$  is a normed space, then the family of seminorms  $p_x(\varphi) = |\varphi(x)|$  for each  $x \in X$  defines the weak\*, or  $w^*$ , topology on  $X^*$ .

The weak and weak\* topologies are very important, and we will discuss them more in depth.

## Metrizability and Normability

When we look at a locally convex space, we concern ourselves with whether the topology of the space is given by a metric or not. It turns out this is (effectively) equivalent to there being countably many seminorms defining the topology.

**Proposition:** Let  $\{p_n : n \geq 1\}$  be a sequence of separating seminorms on  $X$ . For any  $x, y \in X$ , we may take

$$d(x, y) = \sum_{n=1}^{\infty} 2^{-n} \frac{p_n(x - y)}{1 + p_n(x - y)}.$$

Then, the topology on  $X$  is defined by this metric.

**Definition:** A Fréchet space is a locally convex space whose topology is defined by a metric that is complete.

**Definition:** If  $X$  is a topological vector space, and  $B \subseteq X$ , then we say  $B$  is bounded if, for every open set  $U$  containing 0, there is some  $\varepsilon > 0$  such that  $\varepsilon B \subseteq U$ .

**Proposition:** If  $X$  is a locally convex space, then  $X$  is normable if and only if  $X$  has a nonempty bounded open set.

*Proof.* The open unit ball  $U_X$  is bounded. Therefore, we only need to show one direction. Suppose  $X$  is a locally convex space with a bounded open set  $U$ . Translating if necessary, we may assume  $0 \in U$ .

Since  $X$  is locally convex, there is an open, convex, balanced subset of  $U$  which we call  $V$ ; using the gauge of  $V$ , we may find a continuous seminorm  $p$  such that  $\{x : p(x) < 1\} = V$ . We will show that  $p$  is a norm and that  $p$  defines the topology on  $X$ .

First, let  $0 \neq x \in X$ . Let  $W_0$  and  $W_x$  be disjoint open subsets with  $0 \in W_0$  and  $x \in W_x$ . Since  $W_0$  is absorbing, there is some  $\varepsilon > 0$  such that  $W_0 \supseteq \varepsilon U \supseteq \varepsilon V$ . We note that  $\varepsilon V = \{y : p(y) < \varepsilon\}$ . Since  $x \notin W_0$ , we have  $p(x) \geq \varepsilon$ , so that  $p$  is a norm.

To show that the topology on  $X$  is generated by  $p$ , it suffices to show that if  $q$  is any continuous seminorm,

then there is some  $\alpha$  such that  $q \leq \alpha p$ . Since  $q$  is continuous, it follows that there is some  $\varepsilon > 0$  such that  $\{x : q(x) < 1\} \supseteq \varepsilon U \supseteq \varepsilon V$ . Yet, this means that if  $p(x) < \varepsilon$ , then  $q(x) < 1$ , whence  $q \leq \varepsilon^{-1} p$ .  $\square$

## Hyperplane Separation and Continuous Linear Functionals

We recall the classification of continuous linear functionals in a locally convex topological vector space.

**Theorem:** If  $X$  is a locally convex topological vector space with defining seminorms  $\mathcal{P}$ , and  $f : X \rightarrow \mathbb{C}$  is a linear functional, then the following statements are equivalent:

- (i)  $f$  is continuous;
- (ii)  $f$  is continuous at 0;
- (iii)  $f$  is continuous at some point;
- (iv)  $\ker(f)$  is closed;
- (v) there are  $p_1, \dots, p_n \in \mathcal{P}$  and some constant  $C$  such that  $|f(x)| \leq C \max\{p_1(x), \dots, p_n(x)\}$ .

To understand hyperplane separation, we start with the most basic case — namely, that of separating a convex subset from the origin via a continuous linear functional.

**Theorem:** If  $X$  is a topological vector space, and  $G$  is an open convex nonempty subset of  $X$  that does not contain the origin, then there is a closed hyperplane  $M$  such that  $M \cap G = \emptyset$ .

*Proof.* We start by assuming that  $X$  is an  $\mathbb{R}$ -linear space. Choose  $x_0 \in G$ , and let  $H = x_0 - G$ . Then,  $H$  is an open, convex subset of  $X$  that contains 0. Taking the gauge of  $H$  yields a continuous sublinear functional  $q : X \rightarrow \mathbb{R}$  with  $H = \{x : q(x) < 1\}$ . Furthermore, since  $x_0 \notin H$ , we have that  $q(x_0) \geq 1$ .

Consider the subspace  $Y = \{\alpha x_0 : \alpha \in \mathbb{R}\}$ . Let  $f_0 : Y \rightarrow \mathbb{R}$  be defined by taking  $f_0(\alpha x_0) = \alpha q(x_0)$ . Observe that if  $\alpha \geq 0$ , then  $f_0(\alpha x_0) = q(\alpha x_0)$ , and if  $\alpha < 0$ , then  $f_0(\alpha x_0) = \alpha q(x_0) \leq \alpha < 0 \leq q(\alpha x_0)$ , whence  $f_0 \leq q$  on  $Y$ . Using the Hahn–Banach–Minkowski extension yields some  $f : X \rightarrow \mathbb{R}$  a continuous linear functional such that  $f|_Y = f_0$  and  $f \leq q$  on  $X$ . Then,  $M = \ker(f)$  is the desired hyperplane.

If  $x \in G$ , then  $x_0 - x \in H$ , so  $f(x_0) - f(x) = f(x_0 - x) \leq q(x_0 - x) < 1$ , while  $f(x) > f(x_0) - 1 = q(x_0) - 1 \geq 0$  for all  $x \in G$ , whence  $M \cap G = \emptyset$ .

If  $X$  is a  $\mathbb{C}$ -linear space, then since  $X$  is also an  $\mathbb{R}$ -linear space, we obtain a continuous  $\mathbb{R}$ -linear functional  $f : X \rightarrow \mathbb{R}$  with  $G \cap \ker(f) = \emptyset$ . If  $F(x) = f(x) - i f(ix)$ , then  $F$  is a  $\mathbb{C}$ -linear functional with  $f = \operatorname{Re}(F)$  with  $F(x) = 0$  if and only if  $f(x) = f(ix) = 0$ , so that  $M = \ker(F)$  is the desired closed hyperplane.  $\square$

Affine hyperplanes in  $X$  are given by  $M' = M + x_0$  for some  $x_0 \in X$  and hyperplane  $M$ . It turns out that we can extend any separating affine (closed) subspace of a topological vector space to an affine hyperplane.

**Corollary:** Let  $X$  be a topological vector space, and let  $G$  be an open, convex, nonempty subset of  $X$ . If  $Y$  is an affine subspace of  $X$  with  $Y \cap G = \emptyset$ , then there is a closed affine hyperplane  $M$  in  $X$  such that  $Y \subseteq M$  and  $M \cap G = \emptyset$ .

*Proof.* Choosing some  $x_0 \in Y$ , we may consider  $G - x_0$  and  $Y - x_0$ , so that  $Y$  may be assumed to be a subspace of  $X$ . Let  $Q : X \rightarrow X/Y$  be the quotient map. Notice that  $Q^{-1}(Q(G)) = \{y + G : y \in Y\}$ , so that  $Q(G)$  is open in  $X/Y$ . Furthermore,  $Q(G)$  is also convex since  $Q$  is a linear map. Furthermore, since  $Y \cap G = \emptyset$ ,  $0 \notin Q(G)$ . In particular, there is then a closed hyperplane  $N$  in  $X/Y$  such that  $N \cap Q(G) = \emptyset$ . Setting  $M = Q^{-1}(N)$ , we have that  $M$  is our desired hyperplane.  $\square$

In the special case of a real topological vector space, it's useful to note that  $X \setminus \ker(f)$  for a continuous linear functional  $f : X \rightarrow \mathbb{R}$  has two connected components. This yields the following notion.

**Definition:** Let  $X$  be a real topological vector space. A subset  $S \subseteq X$  is called an open half space if there is some continuous linear functional  $f : X \rightarrow \mathbb{R}$  such that  $S = f^{-1}((\alpha, \infty))$ . We call  $S$  a closed half-space if there is a continuous linear functional  $f : X \rightarrow \mathbb{R}$  such that  $S = f^{-1}([\alpha, \infty))$ .

We say  $A$  and  $B$  are strictly separated if they are contained in disjoint open half-spaces, while they are separated if they are contained in two closed half-spaces whose intersection is of the form  $M = f^{-1}(\{\alpha\})$ . That is, their intersection is a closed affine hyperplane. Notice also that continuous linear functionals are open maps, so that the interior of a closed half-space is an open half-space, and the closure of an open

half-space is a closed half-space.

**Theorem** (General Hyperplane Separation): Let  $X$  be a real topological vector space, and  $A$  and  $B$  disjoint convex subsets with  $A$  open. Then, there is a continuous linear functional  $f: X \rightarrow \mathbb{R}$  and a real scalar  $\alpha$  such that  $A \subseteq f^{-1}((-\infty, \alpha))$  and  $B \subseteq f^{-1}([\alpha, \infty))$ .

*Proof.* Let  $G = A - B$ . Then,  $G$  is convex, and  $G = \bigcup_{b \in B} (A - b)$ , meaning  $G$  is open. Since  $A \cap B = \emptyset$ , it follows that  $0 \notin G$ . Thus, there is a closed hyperplane  $M$  in  $X$  such that  $M \cap G = \emptyset$ .

Let  $f: X \rightarrow \mathbb{R}$  be a linear functional such that  $M = \ker(f)$ . Notice that  $f(G)$  is a convex subset of  $\mathbb{R}$  with  $0 \notin f(G)$ . Either,  $f(x) > 0$  for all  $x \in G$  or  $f(x) < 0$  for all  $x \in G$ . Without loss of generality, we may assume that  $f(x) > 0$  for all  $x \in G$ , so that  $f(b) < f(a)$  for any  $a \in A$  and  $b \in B$ , meaning there is some real number  $\alpha$  such that  $\sup_{b \in B} f(b) \leq \alpha \leq \inf_{a \in A} f(a)$ . Since  $f(A)$  and  $f(B)$  are open intervals if  $B$  is open, it follows that  $f < \alpha$  on  $B$  and  $f > \alpha$  on  $A$ .  $\square$

**Lemma:** If  $X$  is a topological vector space,  $K$  is a compact subset of  $X$ , and  $V$  is an open subset satisfying  $K \subseteq V$ , then there is an open neighborhood  $U$  of  $0$  such that  $K + U \subseteq V$ .

*Proof.* Let  $\mathcal{O}_0$  denote the system of open neighborhoods of  $0$ . Suppose toward contradiction that for each  $U \in \mathcal{O}_0$  that  $K + U$  is not contained in  $V$ . We may choose vectors  $x_U \in K$  and  $y_U \in U$  such that  $x_U + y_U \in V^c$ . This yields a net  $(y_U)_{U \in \mathcal{O}_0}$  and  $(x_U)_{U \in \mathcal{O}_0}$  with  $y_U \rightarrow 0$  and a subnet  $x_{U'} \rightarrow x \in K$ . Yet, this yields  $(x_{U'} + y_{U'}) \rightarrow x$ , implying that  $x \in \overline{V^c} = V^c$ , which yields a contradiction.  $\square$

**Theorem** (Hyperplane Separation for Locally Convex Spaces): Let  $X$  be a real locally convex topological vector space, and let  $A$  and  $B$  be two disjoint closed convex subsets of  $X$  with  $B$  compact. Then,  $A$  and  $B$  are contained in disjoint open half-spaces.

*Proof.* Notice that  $B$  is a compact subset of  $A^c$ . Therefore, there is an open neighborhood  $U_1$  of  $0$  such that  $B + U_1 \subseteq A^c$ . Since  $X$  is locally convex, we may find a continuous seminorm  $p$  on  $X$  such that  $\{x : p(x) < 1\} \subseteq U_1$ .

Define  $U = \{x : p(x) < 1/2\}$ . Then,  $(B + U) \cap (A + U) = \emptyset$ , and  $A + U$  and  $B + U$  are open convex subsets of  $X$  that contain  $A$  and  $B$ . Therefore, the result follows from the general hyperplane separation.  $\square$

Rephrasing this theorem for locally convex spaces by using the fact that any linear functional  $\varphi$  over  $\mathbb{C}$  is given by  $\varphi(x) = \operatorname{Re}(\varphi(x)) - i \operatorname{Re}(\varphi(ix))$  yields the following.

**Theorem** (Hyperplane Separation for Locally Convex Spaces, Encore): Let  $X$  be a locally convex space, and let  $A$  and  $B$  be disjoint closed convex subsets of  $X$  with  $B$  compact. Then, there is a continuous linear functional  $f: X \rightarrow \mathbb{C}$ , some  $\alpha \in \mathbb{R}$ , and some  $\varepsilon > 0$  such that for all  $a \in A$  and  $b \in B$ , we have

$$\operatorname{Re}(f(a)) \leq \alpha < \alpha + \varepsilon \leq \operatorname{Re}(f(b)).$$

We will now discuss the dual space of one particular locally convex space.

**Proposition:** Let  $X$  be a locally compact Hausdorff space, and let  $C(X)$  be topologized by uniform convergence on compact subsets. If  $L: C(X) \rightarrow \mathbb{F}$  is a continuous linear functional, then there is a compact set  $K$  and a regular (complex) Borel measure  $\mu$  on  $K$  such that  $L(f) = \int_K f d\mu$  for any  $f \in C(X)$ . Conversely, all these measures define elements of  $C(X)^*$ .

*Proof.* Any compactly support measure defines an element of  $C(X)^*$ , and satisfies  $|L(f)| \leq \|\mu\| \sup_{x \in K} |f(x)|$  for any such measure  $\mu$ .

Conversely, let  $L \in C(X)^*$ . By the classification of continuous linear functionals, there are compact sets  $K_1, \dots, K_n$  and a constant  $C$  such that  $|L(f)| \leq C \max\{p_{K_j}(f) : j = 1, \dots, n\}$ . Let  $K = \bigcup_{j=1}^n K_j$ . Then,  $|L(f)| \leq \alpha p_K(f)$ . In particular, for any  $f \in C(X)$  with  $f|_K = 0$ , then  $L(f) = 0$ .

Define a linear functional  $F: C(K) \rightarrow \mathbb{F}$  as follows. For any  $g \in C(K)$ , let  $\tilde{g}$  be a continuous extension to  $X$ , and set  $F(g) = L(\tilde{g})$ . Suppose  $\tilde{g}_1$  and  $\tilde{g}_2$  are any such extensions; then,  $\tilde{g}_1 - \tilde{g}_2 = 0$  on  $K$ , so that  $L(\tilde{g}_1) = L(\tilde{g}_2)$ , so that  $F$  is well-defined. Furthermore,  $F$  is linear. Since, for any  $g \in C(K)$  and extension  $\tilde{g} \in C(X)$ , we have  $|F(g)| \leq C p_K(g) = C \|g\|$ . In particular, this means  $F$  is continuous, so there is some regular complex Borel measure  $\mu$  in  $M(K)$  such that  $F(g) = \int g d\mu$ . Notice that if  $f \in C(X)$  is such that  $f|_K = g \in C(K)$ ,

it follows that  $L(f) = F(g) = \int f d\mu$ . □

## Weak Topologies and Extremal Structure

### Duality

Let  $X$  be a locally convex space. Then,  $X^*$  denotes the space of continuous linear functionals on  $X$ , and admits a vector space structure. We will write  $\langle x, \varphi \rangle = \varphi(x)$  whenever  $x \in X$  and  $\varphi \in X^*$ . We will write  $\hat{x} = \langle x, \cdot \rangle$  to denote the analogous linear functional on  $\varphi$  such that  $\hat{x}(\varphi) = \langle x, \varphi \rangle$ .

**Definition:** If  $X$  is a locally convex space, then the weak topology on  $X$  is the topology defined by the family of seminorms

$$p_\varphi(x) = |\langle x, \varphi \rangle|.$$

The weak\* topology on  $X$  is the topology defined by the seminorms

$$p_x(\varphi) = |\langle x, \varphi \rangle|.$$

We have that  $(x_i)_i \xrightarrow{\text{wk}} x$  if, for all  $\varphi \in X^*$ ,  $\varphi(x_i) \rightarrow \varphi(x)$ . Similarly, we have that  $(\varphi_i)_i \xrightarrow{\text{wk}^*} \varphi$  if, for all  $x \in X$ ,  $\varphi_i(x) \rightarrow \varphi(x)$ .

**Theorem:** If  $X$  is a locally convex space, then  $(X^*, \text{wk}^*)^* = X$ .

*Proof.* First, if  $x \in X$ , then  $\varphi \mapsto \langle x, \varphi \rangle$  is a weak\*-continuous linear functional on  $X^*$  by definition.

Conversely, if  $f \in (X^*, \text{wk}^*)^*$ , then there are vectors  $x_1, \dots, x_n \in X$  such that

$$|f(\varphi)| \leq \sum_{k=1}^n |\langle x_k, \varphi \rangle|$$

for all  $\varphi \in X^*$ . This implies that

$$\bigcap_{k=1}^n \ker(\hat{x}_k) \subseteq \ker(f),$$

so there are  $\alpha_1, \dots, \alpha_n$  such that  $f = \sum_{k=1}^n \alpha_k \hat{x}_k$ , meaning that  $f \in X$ . □

Local convexity is a very useful property to have, as the following result exhibits.

**Theorem:** Let  $X$  be a locally convex space, and let  $A$  be a convex subset of  $X$ . Then,  $\overline{A} = \overline{A}^{\text{wk}}$ .

*Proof.* Notice that the weak topology is weaker than the topology on  $X$ , so  $\overline{A} \subseteq \overline{A}^{\text{wk}}$ .

Conversely, suppose there were  $x \in \overline{A}^{\text{wk}} \setminus \overline{A}$ . Then, by geometric Hahn–Banach, there is  $\varphi \in X^*$ ,  $\alpha \in \mathbb{R}$ , and  $\varepsilon > 0$  such that

$$\text{Re}(\langle a, \varphi \rangle) \leq \alpha < \alpha + \varepsilon \leq \text{Re}(\langle x, \varphi \rangle)$$

for all  $a \in \overline{A}$ . Yet, this means that  $\overline{A} \subseteq \{y \in X : \text{Re}(\langle y, \varphi \rangle) \leq \alpha\}$ , even though the latter is weakly closed. Since  $x \notin \overline{A}$ , it follows that  $x \notin \overline{A}$ , which yields a contradiction. □

**Corollary:** A convex subset of  $X$  is closed if and only if it is weakly closed.

## Duality in Subspaces and Quotient Spaces

**Proposition:** Let  $p$  be a seminorm on  $X$ ,  $M$  a subspace of  $X$ , and  $\bar{p}: X/M \rightarrow [0, \infty)$  is defined by

$$\bar{p}(x + M) = \inf\{p(x + y) : y \in M\},$$

then  $\bar{p}$  is a seminorm on  $X/M$ . If  $\mathcal{P}$  is a family of continuous seminorms on  $X$  defining a locally convex topology, then  $\bar{\mathcal{P}}$  defines the quotient topology on  $X/M$ .

In particular, if  $X$  is a locally convex space and  $M \leq X$  is a subspace, then  $X/M$  is a locally convex space. Suppose  $f \in (X/M)^*$ . Then,  $f \circ \pi \in X^*$ , where  $\pi: X \rightarrow X/M$  is the projection. Furthermore, observe that  $f \circ \pi$  is an element of

$$M^\perp = \{\varphi \in X^* : \varphi(m) = 0 \text{ for all } m \in M\}.$$

In particular, we obtain the map  $f \mapsto f \circ \pi$ , yielding a linear map  $(X/M)^* \rightarrow M^\perp \subseteq X^*$ .

**Theorem:** Let  $X$  be a locally convex space,  $M \leq X$  a subspace, and  $\pi: X \rightarrow X/M$  the natural map. Then,  $f \mapsto f \circ \pi$  defines a bijection between  $(X/M)^*$  and  $M^\perp$ .

Equipping  $(X/M)^*$  with the weak\* topology, and  $M^\perp$  with the relative weak\* topology, yields that this bijection is a homeomorphism. If  $X$  is a normed space, then the bijection is an isometry and a weak\*-weak\* homeomorphism.

*Proof.* Define  $\rho: (X/M)^* \rightarrow M^\perp$  by  $\rho(f) = f \circ \pi$ . Then, we have that  $\rho$  is well-defined and takes  $(X/M)^*$  to  $M^\perp$ . Notice that if  $0\rho(f)$ , then  $f \circ \pi = 0$ , but since  $\pi$  is surjective,  $f$  must be the zero map, whence  $\rho$  is injective.

Let  $\varphi \in M^\perp$ , and let  $f: X/M \rightarrow \mathbb{C}$  be given by  $f(x + M) = \langle x, \varphi \rangle$ . Since  $M \subseteq \ker(\varphi)$ ,  $f$  is well-defined and linear. Since

$$\pi^{-1}(\{x + M : |f(x + M)| < 1\}) = \{x \in X : |\langle x, \varphi \rangle| < 1\}$$

is open in  $X$ , it follows that  $\{x + M : |f(x + M)| < 1\}$  is open in  $X/M$ , so  $f$  is continuous.

If  $X$  is a normed space, then results from Banach space theory yields that  $\rho$  is an isometry. If  $(f_i)_i \subseteq (X/M)^*$  converges to 0 in the weak\* topology on  $(X/M)^*$ , then for each  $x \in X$ , we have that

$$\begin{aligned} \langle x, \rho(f_i) \rangle &= f_i(\pi(x)) \\ &\rightarrow 0, \end{aligned}$$

so that  $\rho(f_i) \rightarrow 0$  weak\* in  $X^*$ . Conversely, if  $\rho(f_i) \rightarrow 0$  weak\* in  $X^*$ , it follows that for each  $x \in X$ , we have  $f_i(x + M) = \langle x, \rho(f_i) \rangle \rightarrow 0$  weak\* in  $(X/M)^*$ .  $\square$

## Banach-Alaoglu Theorem and Applications

**Theorem:** If  $X$  is a normed space, then the unit ball  $B_{X^*}$  of  $X^*$  is weak\*-compact.

*Proof.* Let  $D_x = \{\alpha \in \mathbb{C} : |\alpha| \leq 1\}$ . Set  $D = \prod_{x \in B_X} D_x$ . By Tychonoff's theorem,  $D$  is compact. Let  $\tau: B_{X^*} \rightarrow D$  be defined by

$$\tau(\varphi)(x) = \langle x, \varphi \rangle.$$

Then,  $\tau(\varphi) \in D$  with  $x$  coordinate given by  $\langle x, \varphi \rangle$ . We will show that  $\tau$  is a homeomorphism from  $B_{X^*}$  with the weak\* topology onto  $\tau(B_{X^*}) \subseteq D$  with the subspace topology, and that  $\tau(B_{X^*}) \subseteq D$  is closed.

Suppose  $\tau(\varphi_1) = \tau(\varphi_2)$ . Then, for each  $x \in B_X$ ,  $\langle x, \varphi_1 \rangle = \langle x, \varphi_2 \rangle$ . Since  $B_X$  separates the points of  $X^*$ , it follows that  $\varphi_1 = \varphi_2$ .

Let  $(\varphi_i)_i$  be a net in  $B_{X^*}$  with  $\varphi_i \xrightarrow{\text{wk}^*} \varphi$ . For each  $x \in B_X$ , it follows that  $\tau(\varphi_i)(x) \rightarrow \tau(\varphi)(x)$ , or that each coordinate of  $(\tau(\varphi_i))_i$  converges to  $\tau(\varphi)$ , so that  $\tau(\varphi_i) \rightarrow \tau(\varphi)$ , meaning  $\tau$  is continuous.

Let  $(\varphi_i)_i$  be a net in  $B_{X^*}$ ,  $f \in D$ , and  $\tau(\varphi_i) \rightarrow f$  in  $D$ . Then,  $f(x) = \lim \langle x, \varphi_i \rangle$  exists for each  $x \in B_X$ . If  $x \in X$ , we choose  $\alpha > 0$  such that  $\|\alpha x\| \leq 1$ . Take  $f(x) = \alpha^{-1} f(\alpha x)$ .

If  $\beta > 0$  is also such that  $\|\beta x\| \leq 1$ , then

$$\begin{aligned} \alpha^{-1} f(\alpha x) &= \alpha^{-1} \lim \langle \alpha x, \varphi_i \rangle \\ &= \beta^{-1} \lim \langle \beta x, \varphi_i \rangle \\ &= \beta^{-1} f(\beta x), \end{aligned}$$

so that  $f(x)$  is well-defined. Notice that by the way  $f$  is defined, it is necessarily a linear functional. If  $\|x\| \leq 1$ , then  $f(x) \in D_x$ , so  $|f(x)| \leq 1$ , meaning that there is  $\varphi \in B_{X^*}$  such that  $\tau(\varphi) = f$ . Thus,  $\tau(B_{X^*})$  is closed in  $D$ , so  $\tau(B_{X^*})$  is compact.  $\square$

Recall that a Banach space  $X$  is called reflexive if the embedding of  $X$  into its double dual  $X^{**}$  is surjective, where we recall that  $\hat{x} \in X^{**}$  is defined by

$$\langle \varphi, \hat{x} \rangle = \langle x, \varphi \rangle.$$

**Theorem** (Goldstine's Theorem): Let  $X$  be a normed space. Then,  $B_X$  is weak\*-dense in  $X^{**}$ .

*Proof.* We will denote by  $w^{**}$  the weak\* topology on  $X^{**}$  induced by  $X^*$ . Let  $B = \overline{B_X}^{w^{**}}$ , and notice that  $B \subseteq B_{X^{**}}$ . Suppose there is some  $\theta \in B_{X^{**}} \setminus B$ . The Hahn–Banach theorem implies there is some  $\varphi \in X^*$ ,  $\alpha \in \mathbb{R}$ , and  $\varepsilon > 0$  such that

$$\operatorname{Re}(\langle x, \varphi \rangle) < \alpha < \alpha + \varepsilon < \operatorname{Re}(\langle \varphi, \theta \rangle)$$

for all  $x \in B_X$ . Since  $0 \in B_X$ , it follows that  $0 < \alpha$ . Dividing, we may assume that there is  $\varphi \in X^*$  and  $\varepsilon > 0$  such that

$$\operatorname{Re}(\langle x, \varphi \rangle) < 1 < 1 + \varepsilon < \operatorname{Re}(\langle \varphi, \theta \rangle).$$

Notice that  $|\langle x, \varphi \rangle| \leq 1$  (multiplying by a phase) for any  $\|x\| \leq 1$ , so that  $\varphi \in B_{X^*}$ . Yet, we also have

$$\begin{aligned} 1 + \varepsilon &< \operatorname{Re}(\langle \varphi, \theta \rangle) \\ &\leq |\langle \varphi, \theta \rangle| \\ &\leq \|\theta\| \\ &\leq 1, \end{aligned}$$

which is a contradiction.  $\square$

**Theorem:** Let  $X$  be a Banach space. The following are equivalent:

- (a)  $X$  is reflexive;
- (b)  $X^*$  is reflexive;
- (c) the weak\* topology on  $X^*$  coincides with the weak topology on  $X^*$  induced by  $X^{**}$ ;
- (d)  $B_X$  is weakly compact.

## $C^*$ -Algebras

### Spectral Theory

Spectral theory is primarily concerned with representations of abelian  $C^*$ -algebras. Consider a normal operator  $N$  on  $H$ . Then,  $C^*(N)$  is an abelian  $C^*$ -algebra, and the continuous functional calculus  $f \mapsto f(N)$  yields a  $*$ -isomorphism of  $C(\sigma(N))$  onto  $C^*(N)$ . In particular,  $f \mapsto f(N)$  is a representation of the abelian  $C^*$ -algebra  $C^*(N)$ .

We will find a way to extend this further, as well as find some useful applications to the structure of abelian von Neumann algebras and understanding unitary invariants for normal operators.

**Definition:** Let  $X$  be a set,  $\Sigma$  a  $\sigma$ -algebra on  $X$ , and  $H$  a Hilbert space. A *spectral measure* for  $(X, \Sigma, H)$  is a map  $E: \Sigma \rightarrow B(H)$  satisfying

- (a) for each  $\Delta \in \Sigma$ ,  $E(\Delta)$  is a projection;
- (b)  $E(\emptyset) = 0$  and  $E(X) = 1$ ;
- (c)  $E(\Delta_1 \cap \Delta_2) = E(\Delta_1)E(\Delta_2)$  for any  $\Delta_1, \Delta_2 \in \Sigma$ ;
- (d) given pairwise disjoint  $\{\Delta_n\}_{n=1}^\infty$ , we have

$$E\left(\bigcup_{n=1}^\infty \Delta_n\right) = \sum_{n=1}^\infty E(\Delta_n),$$

where the latter is an SOT-convergent sum of pairwise orthogonal projections converging to the projection onto the supremum  $\bigwedge_{n=1}^\infty E(\Delta_n)H$ .

Recall that the SOT and WOT are defined by the seminorms

- $p_h(A) = \|Ah\|$  for the SOT, and
- $p_{h,k}(H) = |\langle Ah, k \rangle|$  for the WOT.

**Example:** If  $(X, \Omega, \mu)$  is a  $\sigma$ -finite measure space, and  $\phi \in L^\infty(X, \mu)$ , then  $M_\phi \in B(L^2(X, \mu))$  can be defined by  $M_\phi(f) = \phi f$ .

A net  $(\phi_i)_{i \in I}$  in  $L^\infty(X, \mu)$  is  $w^*$ -convergent to  $\phi$  if and only if  $M_{\phi_i} \rightarrow M_\phi$  in WOT. This follows from the fact that if  $f, g \in L^2(X, \mu)$ , then  $f\bar{g} \in L^1(X, \mu)$ , where we have  $L^1(X, \mu)^* = L^\infty(X, \mu)$ , so that if  $(\phi_i)_i \rightarrow \phi$  weak\*, then

$$\begin{aligned} \langle M_{\phi_i} f, g \rangle &= \int \phi_i(f\bar{g}) d\mu \\ &\rightarrow \int \phi f\bar{g} d\mu \\ &= \langle M_\phi f, g \rangle, \end{aligned}$$

while the converse emerges from the fact that any element  $f \in L^1(X, \mu)$  can be written as  $f = g_1 \bar{g}_2$ .

**Example:** If  $E$  is a spectral measure for  $(X, \Omega, H)$ , then  $E^{(n)}(\Delta) = E(\Delta)^{(n)}$  (under the diagonal embedding) is a spectral measure for  $(X, \Omega, H^{(n)})$ .

We will now use spectral measures to define a representation of a  $C^*$ -algebra.

**Lemma:** Let  $E$  be a spectral measure on  $(X, \Omega, H)$ , and let  $g, h \in H$ . Then,

$$E_{g,h}(\Delta) = \langle E(\Delta)g, h \rangle$$

defines a countably additive measure on  $\Omega$  with total variation bounded above by  $\|g\|\|h\|$ .

*Proof.* For countable additivity, we note the classic result that a sequence of projections  $(P_n)_n \rightarrow P$  converges in WOT if and only if it converges in SOT. In particular, this means that if  $\{\Delta_n\}_{n \in \mathbb{N}}$  is a sequence of disjoint subsets of  $X$ , then we have a sequence of projections

$$\left( \sum_{k=0}^n E(\Delta_k) \right)_n \rightarrow \sum_{k=0}^\infty E(\Delta_k)$$

converging in SOT, hence converging in WOT. Yet, this means that

$$\begin{aligned} \left\langle \left( \sum_{k=0}^n E(\Delta_k) \right) g, h \right\rangle &= \sum_{k=0}^n \langle E(\Delta_k) g, h \rangle \\ &\rightarrow \left\langle \left( \sum_{k=0}^{\infty} E(\Delta_k) \right) g, h \right\rangle, \end{aligned}$$

so that

$$\sum_{k=0}^{\infty} \langle E(\Delta_k) g, h \rangle = \left\langle E \left( \bigcup_{k=0}^{\infty} \Delta_k \right) g, h \right\rangle,$$

yielding our desired result.

If we have a finite set  $\Delta_1, \dots, \Delta_n$  of pairwise disjoint sets, we choose  $|\alpha_j| = 1$  such that  $\left| \langle E(\Delta_j) g, h \rangle \right| = \alpha_j \langle E(\Delta_j) g, h \rangle$ , whence

$$\begin{aligned} \sum_{j=1}^n |\mu(\Delta_j)| &= \left\langle \sum_{j=1}^n \alpha_j E(\Delta_j) g, h \right\rangle \\ &\leq \left\| \sum_{j=1}^n E(\Delta_j) \alpha_j g \right\| \|h\|. \end{aligned}$$

Notice that the family of  $E(\Delta_j)$  are pairwise orthogonal projections, meaning that by Pythagoras's Theorem and the additivity of the spectral measure, we have

$$\begin{aligned} \left\| \sum_{j=1}^n E(\Delta_j) \alpha_j g \right\|^2 &= \sum_{j=1}^n \|E(\Delta_j) g\|^2 \\ &= \left\| E \left( \bigcup_{j=1}^n \Delta_j \right) g \right\|^2 \\ &\leq \|g\|^2, \end{aligned}$$

whence  $\sum_{j=1}^n |\mu(\Delta_j)| \leq \|g\| \|h\|$ . Since this holds for any finite collection of subsets, we have  $\|\mu\|_{\text{TV}} \leq \|g\| \|h\|$ .  $\square$

We next concern ourselves with establishing “integration” with respect to the spectral measure.

**Proposition:** Let  $E$  be a spectral measure for  $(X, \Omega, H)$ , and let  $\phi: X \rightarrow \mathbb{C}$  be a bounded  $\Omega$ -measurable function. Then, there is a unique operator  $A \in B(H)$  such that for any  $\varepsilon > 0$  and partition  $\{\Delta_1, \dots, \Delta_n\}$  with  $\sup_{x, x' \in \Delta_k} |\phi(x) - \phi(x')| < \varepsilon$  for each  $k$ , we have

$$\left\| A - \sum_{k=1}^n \phi(x_k) E(\Delta_k) \right\| < \varepsilon$$

for any  $x_k \in \Delta_k$ .

*Proof.* For each  $g, h \in H$ , define a (bounded) sesquilinear form

$$B(g, h) = \int \phi dE_{g, h},$$

with  $|B(g, h)| \leq \|\phi\|_{L^\infty} \|g\| \|h\|$ . Then, by the structure of bounded sesquilinear forms, there is a unique operator  $A$  such that  $B(g, h) = \langle Ag, h \rangle$  for all  $g, h \in H$ .

Consider a partition  $\{\Delta_1, \dots, \Delta_n\}$  satisfying the conditions in the proposition. For any arbitrary vectors  $g, h \in H$  and  $x_k \in \Delta_k$ , we have

$$\begin{aligned} \left| \langle Ag, h \rangle - \sum_{k=1}^n \phi(x_k) \langle E(\Delta_k)g, h \rangle \right| &= \left| \sum_{k=1}^n \int_{\Delta_k} (\phi(x) - \phi(x_k)) dE_{g,h} \right| \\ &\leq \sum_{k=1}^n \int_{\Delta_k} |\phi(x) - \phi(x_k)| d|E_{g,h}| \\ &\leq \varepsilon \int d|E_{g,h}| \\ &\leq \varepsilon \|g\| \|h\|. \end{aligned}$$

This yields the desired result. □

For the operator  $A$  in the above statement, we write

$$A = \int \phi dE.$$

That is, for any  $g, h \in H$  and bounded  $\Omega$ -measurable function  $\phi$  on  $X$ , we have

$$\left\langle \left( \int \phi dE \right) g, h \right\rangle = \int \phi dE_{g,h}.$$

If  $B_\infty(X, \Omega)$  denotes the set of bounded  $\Omega$ -measurable functions  $\phi: X \rightarrow \mathbb{C}$  with the norm  $\|\phi\| = \sup_{x \in X} |\phi(x)|$ , then  $B_\infty(X, \Omega)$  is a unital Banach algebra — in fact, it is an abelian  $C^*$ -algebra.

**Proposition:** Suppose  $E$  is a spectral measure for  $(X, \Omega, H)$ , and  $\rho: B_\infty(X, \Omega) \rightarrow B(H)$  is defined by taking  $\rho(\phi) = \int \phi dE$ .

Then,  $\rho$  is a representation of  $B_\infty(X, \Omega)$  and  $\rho(\phi)$  is a normal operator for every  $\phi \in B_\infty(X, \Omega)$ .

*Proof.* We will only focus on showing that  $\rho$  is multiplicative. Suppose  $\phi, \psi \in B_\infty(X, \Omega)$ . Let  $\varepsilon > 0$ , and let  $\{\Delta_1, \dots, \Delta_n\}$  be a Borel partition of  $X$  satisfying

$$\sup \{ |\omega(x) - \omega(x')| : x, x' \in \Delta_k, \omega = \phi, \psi, \phi\psi \} < \varepsilon.$$

In particular, we have that for any  $x_k \in \Delta_k$  and  $\omega = \phi, \psi, \phi\psi$ , we have

$$\left\| \int \omega dE - \sum_{k=1}^n \omega(x_k) E(\Delta_k) \right\| < \varepsilon.$$

The triangle inequality then gives

$$\begin{aligned} \left\| \int \phi\psi dE - \left( \int \phi dE \right) \left( \int \psi dE \right) \right\| &\leq \varepsilon + \left\| \sum_{k=1}^n \phi(x_k) \psi(x_k) E(\Delta_k) - \left( \sum_{i=1}^n \phi(x_i) E(\Delta_i) \right) \left( \sum_{j=1}^n \psi(x_j) E(\Delta_j) \right) \right\| \\ &\quad + \left\| \left( \sum_{i=1}^n \phi(x_i) E(\Delta_i) \right) \left( \sum_{j=1}^n \psi(x_j) E(\Delta_j) \right) - \left( \int \phi dE \right) \left( \int \psi dE \right) \right\|. \end{aligned}$$

Since  $E(\Delta_i)E(\Delta_j) = E(\Delta_i \cap \Delta_j)$ , and  $\{\Delta_1, \dots, \Delta_n\}$  is a partition, we have that the second term on the right-hand side is zero. Thus, we have

$$\left\| \int \phi\psi dE - \left( \int \phi dE \right) \left( \int \psi dE \right) \right\| \leq \varepsilon + \left\| \left( \sum_{i=1}^n \phi(x_i) E(\Delta_i) \right) \left( \sum_{j=1}^n \psi(x_j) E(\Delta_j) - \int \psi dE \right) \right\|$$

$$+ \left\| \left( \sum_{i=1}^n \phi(x_i) E(\Delta_i) - \int \phi dE \right) \left( \int \psi dE \right) \right\|$$

$$\leq \varepsilon (1 + \|\phi\| + \|\psi\|).$$

Thus, we get multiplicativity.  $\square$

**Corollary:** If  $X$  is a compact Hausdorff space, and  $E$  is a spectral measure on  $H$  defined on the Borel sets of  $X$ , then  $\rho: C(X) \rightarrow B(H)$  given by  $\rho(u) = \int u dE$  is a representation of  $C(X)$ .

Thus far, all we have obtained is a variety of the properties of a spectral measure. However, we do not know if such a measure exists. The following is the proof that it does.

**Theorem:** Suppose  $\rho: C(X) \rightarrow B(H)$  is a representation. Then, there is a unique spectral measure  $E$  defined on the Borel subsets of  $X$  such that for all  $g, h \in H$ ,  $E_{g,h}$  is a regular measure, and we have

$$\rho(u) = \int u dE$$

for all  $u \in C(X)$ .

**Theorem:** Let  $g, h \in H$ . The map  $u \mapsto \langle \rho(u)g, h \rangle$  is a linear functional on  $C(X)$  with norm at most  $\|g\|\|h\|$ . So, there is a unique measure  $\mu_{g,h}$  in  $M(X)$  such that

$$\langle \rho(u)g, h \rangle = \int u d\mu_{g,h}.$$

The pairing  $(g, h) \mapsto \mu_{g,h}$  is sesquilinear, and for fixed  $\phi \in B_\infty(X)$ , the form

$$[g, h] = \int \phi d\mu_{g,h}$$

is a sesquilinear form with  $\|[g, h]\| \leq \|\phi\| \|g\| \|h\|$ . Therefore, there is a unique bounded operator  $A$  such that  $[g, h] = \langle Ag, h \rangle$  with  $\|A\| \leq \|\phi\|$ . We will denote this operator  $A$  by  $\tilde{\rho}(\phi)$ . We obtain a well-defined map  $\tilde{\rho}: B_\infty(X) \rightarrow B(H)$  satisfying  $\|\tilde{\rho}(\phi)\| \leq \|\phi\|$  such that

$$\langle \tilde{\rho}(\phi)g, h \rangle = \int \phi d\mu_{g,h}.$$

We start by claiming that  $\tilde{\rho}: B_\infty(X) \rightarrow B(H)$  is a representation with  $\tilde{\rho}|_{C(X)} = \rho$ . That  $\tilde{\rho}$  restricted to  $C(X)$  is equal to  $\rho$  follows directly from the definition of  $\tilde{\rho}$ .

Now, let  $\phi \in B_\infty(X)$ . Consider  $\phi$  as an element of  $M(X)^* = C(X)^{**}$  by viewing it as corresponding to the linear functional

$$\mu \mapsto \int \phi d\mu.$$

By the Banach–Alaoglu theorem, we have that  $\{u \in C(X) : \|u\| \leq \|\phi\|\}$  is weak\*-dense in  $\{L \in M(X)^* : \|L\| \leq \|\phi\|\}$ . Then, there is a net  $(u_i)_i \subseteq C(X)$  with  $\|u_i\| \leq \|\phi\|$  for all  $i$  and

$$\int \phi d\mu = \lim_i \int u_i d\mu$$

for all  $\mu \in M(X)$ . For any  $\psi \in B_\infty(X)$ , we have  $\psi\mu \in M(X)$  for any  $\mu \in M(X)$ . In particular, we have

$$\int \phi\psi d\mu = \lim_i \int u_i\psi d\mu$$

for all  $\psi \in B_\infty(X)$  and  $\mu \in M(X)$ , whence  $\tilde{\rho}(u_i\psi) \rightarrow \tilde{\rho}(\phi\psi)$  in WOT. For any  $\psi \in C(X)$ , we have that

$$\begin{aligned}\tilde{\rho}(\phi\psi) &= \lim_i \tilde{\rho}(u_i\psi) \\ &= \lim_i \rho(u_i)\rho(\psi) \\ &= \tilde{\rho}(\phi)\rho(\psi).\end{aligned}$$

In particular, we have  $\tilde{\rho}(u_i\psi) = \rho(u_i)\tilde{\rho}(\psi)$  for any  $\psi \in B_\infty(X)$  for all  $u_i \in C(X)$ . Therefore, we have  $\tilde{\rho}(u_i\psi) \rightarrow \tilde{\rho}(\phi\psi)$  in WOT, whence

$$\tilde{\rho}(\phi\psi) = \tilde{\rho}(\phi)\tilde{\rho}(\psi)$$

for any  $\phi, \psi \in B_\infty(X)$ .

The linearity of measures gives that  $\tilde{\rho}$  itself is linear. A similar bootstrapping argument gives  $\tilde{\rho}(\phi)^* = \tilde{\rho}(\bar{\phi})$ .

Now, let  $\Delta \subseteq X$  be Borel. Let  $E(\Delta) = \tilde{\rho}(\chi_\Delta)$ . We will show that  $E$  is a spectral measure.

First, since  $\chi_\Delta$  is a hermitian idempotent in  $B_\infty(X)$ , it follows that  $E(\Delta)$  is a projection by the classification of projections on a Hilbert space. Furthermore,  $\chi_\emptyset = 0$  and  $\chi_X = 1$ , so that  $E(\emptyset) = 0$  and  $E(X) = 1$ . Since

$$\begin{aligned}E(\Delta_1 \cap \Delta_2) &= \tilde{\rho}(\chi_{\Delta_1 \cap \Delta_2}) \\ &= \tilde{\rho}(\chi_{\Delta_1} \chi_{\Delta_2}) \\ &= E(\Delta_1)E(\Delta_2).\end{aligned}$$

Finally, let  $\{\Delta_n : n \geq 0\}$  be a pairwise disjoint sequence of Borel sets. Then, for fixed  $h \in H$  and  $\Lambda_n = \bigcup_{k=n+1}^\infty \Delta_k$ , we have

$$\begin{aligned}\left\| E\left(\bigcup_{k=1}^\infty \Delta_k\right) - \sum_{k=1}^n E(\Delta_k) \right\|^2 &= \langle E(\Lambda_n)h, E(\Lambda_n)h \rangle \\ &= \langle E(\Lambda_n)h, h \rangle \\ &= \int \chi_{\Lambda_n} d\mu_{h,h} \\ &= \sum_{k=n+1}^\infty \mu_{h,h}(\Delta_k) \\ &\xrightarrow{n \rightarrow \infty} 0.\end{aligned}$$

Thus,  $E$  is a spectral measure.

Finally, we must show that  $\rho(u) = \int u dE$  for all  $u \in C(X)$ . In fact, we will show that  $\tilde{\rho}(\phi) = \int \phi dE$  for all  $\phi \in B_\infty(X)$ . For a fixed  $\phi \in B_\infty(X)$ , we choose a Borel partition  $\{\Delta_1, \dots, \Delta_n\}$  such that  $\sup\{|\phi(x) - \phi(x')| : x, x' \in \Delta_k\} < \varepsilon$  for each  $k$ . Then,

$$\left\| \phi - \sum_{k=1}^n \phi(x_k) \chi_{\Delta_k} \right\| < \varepsilon$$

for any choices of  $x_k \in \Delta_k$ . Since  $\|\tilde{\rho}\| = 1$ , it follows that

$$\left\| \tilde{\rho}(\phi) - \sum_{k=1}^n \phi(x_k) E(\Delta_k) \right\| < \varepsilon,$$

so that  $\tilde{\rho}(\phi) = \int \phi dE$  for any  $\phi \in B_\infty(X)$ .

The uniqueness of  $E$  follows from the fact that the measures  $E_{g,h}$  are unique for each  $g, h \in H$  as part of the Riesz Representation Theorem.

The primary use of the existence of spectral measures is to enable the Borel functional calculus for normal operators on Hilbert spaces.

**Theorem:** Let  $N$  be a normal operator on  $H$ . There is a unique spectral measure  $E$  on the Borel subsets of  $\sigma(N)$  such that:

- (a)  $N = \int z dE$ ;
- (b) if  $G$  is a nonempty open subset of  $\sigma(N)$ , then  $E(G) \neq 0$ ;
- (c) if  $T \in B(H)$ , then  $TN = NT$  and  $TN^* = N^*T$  if and only if  $TE(\Delta) = E(\Delta)T$  for all Borel  $\Delta \subseteq \sigma(N)$ .

*Proof.* Let  $A = C^*(N)$ . Then,  $A$  is the closure of the polynomials in  $N$  and  $N^*$ , so there is an isometric  $*$ -isomorphism  $\rho: C(\sigma(N)) \rightarrow A$  given by the functional calculus,  $\rho(u) = u(N)$ . There is a unique spectral measure  $E$  defined on the Borel subsets of  $\sigma(N)$  such that  $\rho(u) = \int u dE$  for all  $u \in C(\sigma(N))$ . This gives (a).

If  $G$  is a nonempty open subset of  $\sigma(N)$ , Urysohn's Lemma yields a continuous function  $u$  on  $\sigma(N)$  such that  $0 \leq u(z) \leq \chi_G(z)$  for all  $z \in \sigma(N)$ . Since  $E(G) = \overline{\rho(\chi_G)} \geq \rho(u) \neq 0$ , it follows that  $E(G) \neq 0$ .

Let  $T \in B(H)$  be such that  $TN = NT$  and  $TN^* = N^*T$ . Then, by the Stone–Weierstrass theorem, it follows that  $Tu(N) = u(N)T$  for all  $u \in C(\sigma(N))$ .

Consider the set  $\Omega$  defined as follows:

$$\Omega = \{\Delta \subseteq \sigma(N) : \Delta \text{ is a Borel subset and } AE(\Delta) = E(\Delta)A\}.$$

Notice that by the definition of the spectral measure,  $\Omega$  is closed under finite intersections, countable unions (using SOT convergence), and contains  $\sigma(N)$ , so it is a  $\sigma$ -algebra by the Dynkin  $\pi$ - $\lambda$  theorem. If  $G \subseteq \sigma(N)$  is open, there is a sequence  $(u_n)_n$  of positive continuous functions on  $\sigma(N)$  such that  $u_n(z) \nearrow \chi_G(z)$  for all  $z \in \sigma(N)$ . Dominated convergence then gives

$$\begin{aligned} \langle AE(G)g, h \rangle &= \langle E(G)g, A^*h \rangle \\ &= E_{g, A^*h}(G) \\ &= \int \chi_G(z) dE_{g, A^*h} \\ &= \int \lim_{n \rightarrow \infty} u_n(z) dE_{g, A^*h} \\ &= \lim_{n \rightarrow \infty} \int u_n(z) dE_{g, A^*h} \\ &= \lim_{n \rightarrow \infty} \langle Au_n(N)g, h \rangle \\ &= \lim_{n \rightarrow \infty} \langle u_n(N)Ag, h \rangle \\ &= \langle E(G)Ag, h \rangle. \end{aligned}$$

Thus,  $\Omega$  contains all the open subsets of  $\sigma(N)$ , so it must be the collection of Borel sets. The converse follows from bootstrapping by approximating with simple functions.  $\square$

The spectral measure will provide us a way to reframe normal operators as multiplication operators on  $L^\infty(X, \mu)$  for some measure space  $(X, \mu)$ .

**Definition:** We call  $e_0 \in H$  *star-cyclic* for an operator  $A$  if the smallest reducing subspace for  $A$  that contains  $e_0$  is  $H$ .

The operator  $A$  is *star-cyclic* if it contains a star-cyclic vector.

Similarly,  $e_0 \in H$  is *cyclic* for  $A$  if  $H$  is the smallest invariant subspace for  $A$  that contains  $e_0$ , and  $A$  is *cyclic* if it contains a cyclic vector.

Equivalently,  $e_0$  is star-cyclic if and only if

$$H = \overline{\{Te_0 : T \in C^*(A)\}}$$

and  $e_0$  is cyclic if and only if

$$H = \overline{\{p(A)e_0 : p \in \mathbb{C}[x]\}}.$$

**Proposition:** If  $A$  has either a cyclic or star-cyclic vector, then  $H$  is separable.

*Proof.* Notice that  $C^*(A)$  and  $\{p(A) : p \in \mathbb{C}[x]\}$  are separable subalgebras of  $B(H)$ .  $\square$

If  $\mu$  is a compactly supported measure on  $K \subseteq \mathbb{C}$ , and  $N_\mu$  is the operator defined by  $N_\mu \equiv M_z$ , where  $(M_z f)(z) = zf(z)$ , then we observe that  $C^*(N_\mu)$  is given by  $M_u$  for all  $u \in C(K)$ . Since continuous functions are dense in  $L^2(\mu)$ , it follows that 1 is a star-cyclic vector for  $N_\mu$ .

Furthermore, the converse holds.

**Theorem:** A normal operator  $N$  is star-cyclic if and only if  $N$  is unitarily equivalent to  $N_\mu$  for some compactly supported measure  $\mu$  on  $\mathbb{C}$ . If  $e_0$  is a star-cyclic vector for  $N$ , then  $\mu$  can be chosen such that there is an isomorphism  $V : H \rightarrow L^2(\mu)$  with  $Ve_0 = 1$  and  $VNV^{-1} = N_\mu$ . Furthermore,  $V$  is unique.

*Proof.* Suppose  $N \cong N_\mu$ . Then, we have already seen that  $N$  is star-cyclic.

Suppose  $N$  has a star-cyclic vector  $e_0$ . Let  $E$  be the spectral measure for  $N$ . Set  $\mu(\Delta) = \langle E(\Delta)e_0, e_0 \rangle = \|E(\Delta)e_0\|^2$  for each Borel subset  $\Delta$  of  $\mathbb{C}$ .

Let  $K = \text{supp}(\mu)$ . For any  $\phi \in B_\infty(K)$ , we have

$$\begin{aligned} \|\phi(N)e_0\|^2 &= \langle |\phi|^2(N)e_0, e_0 \rangle \\ &= \int |\phi(z)|^2 d\langle E(z)e_0, e_0 \rangle \\ &= \int |\phi|^2 d\mu. \end{aligned}$$

Viewing  $B_\infty(K) \subseteq L^2(\mu)$ , we have that  $U\phi = \phi(N)e_0$  defines an isometry from  $B_\infty(K)$  onto  $\{\phi(N)e_0 : \phi \in B_\infty(K)\}$ .

Since  $e_0$  is star-cyclic, it follows that the range of  $U$  is dense in  $H$ , so  $U$  extends to an isomorphism  $U : L^2(\mu) \rightarrow H$ .

For any  $\phi \in B(K)$ , then we have

$$\begin{aligned} UN_\mu U^{-1}(\phi(N)e_0) &= UN_\mu(\phi) \\ &= U(z\phi) \\ &= N\phi(N)e_0, \end{aligned}$$

so that  $UN_\mu U^{-1} = N$  on the dense subset  $\{\phi(N)e_0 : \phi \in B_\infty(K)\}$  of  $L^2(\mu)$ . Therefore  $UN_\mu U^{-1} = N$ , and  $V = U^{-1}$  is our desired selection.  $\square$

We say two measures  $\mu_1$  and  $\mu_2$  are mutually absolutely continuous (or of the same measure class) if they have the same null sets. We will write  $[\mu_1] = [\mu_2]$ . If the respective Radon–Nikodym derivatives  $\frac{d\mu_1}{d\mu_2}$  and  $\frac{d\mu_2}{d\mu_1}$  are essentially bounded, then we say  $\mu_1$  and  $\mu_2$  are boundedly mutually absolutely continuous.

**Theorem:** We have that  $N_{\mu_1} \cong N_{\mu_2}$  if and only if  $[\mu_1] = [\mu_2]$ .

*Proof.* Suppose  $[\mu_1] = [\mu_2]$ , and let  $\phi = \frac{d\mu_1}{d\mu_2}$ . If  $g \in L^1(\mu_1)$ , then  $g\phi \in L^1(\mu_2)$ , with

$$\int g\phi d\mu_2 = \int g d\mu_1.$$

For any  $f \in L^2(\mu_1)$ , we have  $f\sqrt{\phi} \in L^2(\mu_2)$  with

$$\|f\sqrt{\phi}\|_{L^2(\mu_2)} = \|f\|_{L^2(\mu_1)}.$$

Define  $U: L^2(\mu_1) \rightarrow L^2(\mu_2)$  by  $Uf = f\sqrt{\phi}$ . This is an isometry.

For any  $g \in L^2(\mu_2)$ , we have that  $f = \phi^{-1/2}g \in L^2(\mu_1)$  with  $Uf = g$ , so  $U$  is surjective with  $U^{-1}g = \phi^{-1/2}g$  for any  $g \in L^2(\mu_2)$ . For any  $g \in L^2(\mu_2)$ ,  $UN_{\mu_1}U^{-1}g = zg$ , so  $UN_{\mu_1}U^{-1} = N_{\mu_2}$ .

Now, suppose  $V: L^2(\mu_1) \rightarrow L^2(\mu_2)$  is an isomorphism satisfying  $VN_{\mu_1}V^{-1} = N_{\mu_2}$ . Let  $\psi = V(1)$ , with  $\psi \in L^2(\mu_2)$ . We will write  $N_j = N_{\mu_j}$ . Notice that for any polynomial  $p \in \mathbb{C}[z, \bar{z}]$ , we have  $Vp(N_1, N_1^*)V^{-1} = p(N_2, N_2^*)$ . Since  $N_1 \cong N_2$ ,  $\sigma(N_1) = \sigma(N_2)$ , so  $\mu_1$  and  $\mu_2$  have the same support, which we will dub  $K$ .

Taking uniform limits of polynomials yields for all  $u \in C(K)$ , we have  $Vu(N_1)V^{-1} = u(N_2)$ , so we have  $V(u) = u\psi$ . Since  $V$  is an isometry, we have

$$\int |u|^2 d\mu_1 = \int |u|^2 |\psi|^2 d\mu_2$$

for every  $u \in C(K)$ . Therefore, we have

$$\int v d\mu_1 = \int v |\psi|^2 d\mu_2$$

for all  $v \in C(K)$  with  $v \geq 0$ . By the Riesz representation theorem, we have that  $\mu_1 = |\psi|^2 \mu_2$ , so  $\mu_1 \ll \mu_2$ . Reversing the roles of  $N_1$  and  $N_2$ , and substituting  $V^{-1}$  for  $V$  where necessary, we obtain that  $\mu_2 \ll \mu_1$ , whence  $[\mu_1] = [\mu_2]$ .  $\square$

## Applications of the Spectral Theorem

**Proposition:** If  $N$  is a normal operator with spectral decomposition  $N = \int z dE(z)$ , we have that  $N$  is compact if and only if for every  $\varepsilon > 0$ ,  $E(\{z : |z| > \varepsilon\})$  is finite rank.

*Proof.* Let  $\varepsilon > 0$ , and let  $\Delta_\varepsilon = \{z : |z| > \varepsilon\}$ , with  $E_\varepsilon = E(\Delta_\varepsilon)$ . Then,

$$\begin{aligned} N - NE_\varepsilon &= \int z \chi_{\Delta_\varepsilon^c} dE(z) \\ &= \phi(N), \end{aligned}$$

where  $\phi(z) = z \chi_{\Delta_\varepsilon^c}(z)$ . In particular,  $\|N - NE_\varepsilon\| \leq \varepsilon$ . If  $E_\varepsilon$  has finite rank for every  $\varepsilon > 0$ , then  $NE_\varepsilon$  has finite rank, so  $N \in \overline{F(H)}$ .

If  $N$  is compact, then  $\phi(z) = z^{-1} \chi_{\Delta_\varepsilon}(z)$  has  $\phi(z) \in B(\mathbb{C})$ . Since  $N$  is compact, we have that  $N\phi(N)$  is compact, but then

$$\begin{aligned} N\phi(N) &= \int zz^{-1} \chi_{\Delta_\varepsilon}(z) dE(z) \\ &= E_\varepsilon, \end{aligned}$$

so  $E_\varepsilon$  is a compact projection, whence it is finite rank.  $\square$

**Theorem:** If  $H$  is separable, and  $I$  is an ideal of  $B(H)$  that contains a noncompact operator, then  $I = B(H)$ .

*Proof.* Let  $A \in I \setminus K(H)$ . Take the spectral decomposition

$$A^*A = \int t dE(t).$$

By the preceding proposition, there is some  $\varepsilon > 0$  such that  $P = E(\{t \geq 0 : \varepsilon < t < \infty\})$  has infinite rank. This yields that

$$P = \left( \int t^{-1} \chi_{(\varepsilon, \infty)}(t) dE(t) \right) A^*A$$

$\in I$ .

Since  $H$  is separable,  $\text{hdim}(PH) = \aleph_0 = \text{hdim}(H)$ . There is a surjective isometry  $U: H \rightarrow PH$  satisfying  $1 = U^*PU$ . Since  $P \in I$ ,  $1 \in I$ , so  $I = B(H)$ .  $\square$

## References

- [Ped89] Gert K. Pedersen. *Analysis Now*. Vol. 118. Graduate Texts in Mathematics. Springer-Verlag, New York, 1989, pp. xiv+277. ISBN: 0-387-96788-5. DOI: [10.1007/978-1-4612-1007-8](https://doi.org/10.1007/978-1-4612-1007-8). URL: <https://doi.org/10.1007/978-1-4612-1007-8>.
- [Con90] John B. Conway. *A Course in Functional Analysis*. Second. Vol. 96. Graduate Texts in Mathematics. Springer-Verlag, New York, 1990, pp. xvi+399. ISBN: 0-387-97245-5.